Introduction to Functions in Java

1. What are Functions?

A function (also called a method in Java) is a block of code that performs a specific task. Functions allow you to group related code together and reuse it whenever needed.

Basic Structure of a Function in Java

```java
public static returnType functionName(parameters) {
    // code to be executed
    return value; // if returnType is not void
}
```

Components of a Function:

- **Access Modifier**: `public`, `private`, `protected`
- **Static Keyword**: Makes the function belong to the class rather than an instance
- **Return Type**: The data type of the value the function returns (`int`, `String`, `void`, etc.)
- **Function Name**: A descriptive name for the function
- **Parameters**: Input values the function accepts (optional)
- **Function Body**: The code that executes when the function is called

Example of a Simple Function

```java
public static void greetUser() {
    System.out.println("Hello, welcome to Java programming!");
}
```

Function with Parameters and Return Value

```java
public static int addNumbers(int a, int b) {
    int sum = a + b;
    return sum;
}
```

Function with Different Return Types

```java
// Function returning a String
public static String getFullName(String firstName, String lastName) {
    return firstName + " " + lastName;
}

// Function returning void (no return value)
public static void displayMessage(String message) {
    System.out.println("Message: " + message);
}

// Function returning a boolean
public static boolean isEven(int number) {
    return number % 2 == 0;
}
```

2. Why We Need Functions

2.1 Code Reusability

Functions allow you to write code once and use it multiple times, avoiding repetition.

Without Functions:

```java
public class Main {
    public static void main(String[] args) {
        // Calculating area of rectangle 1
        int length1 = 5;
        int width1 = 3;
        int area1 = length1 * width1;
        System.out.println("Area of rectangle 1: " + area1);

        // Calculating area of rectangle 2
        int length2 = 8;
        int width2 = 4;
        int area2 = length2 * width2;
        System.out.println("Area of rectangle 2: " + area2);
    }
}
```

With Functions:

```java
public class Main {
    public static int calculateArea(int length, int width) {
        return length * width;
    }

    public static void main(String[] args) {
        int area1 = calculateArea(5, 3);
        int area2 = calculateArea(8, 4);

        System.out.println("Area of rectangle 1: " + area1);
        System.out.println("Area of rectangle 2: " + area2);
    }
}
```

2.2 Better Organization

Functions help organize code into logical sections, making it easier to read and understand.

```java
public class Calculator {
    public static int add(int a, int b) {
        return a + b;
    }

    public static int subtract(int a, int b) {
        return a - b;
    }

    public static int multiply(int a, int b) {
        return a * b;
    }

    public static double divide(int a, int b) {
        if (b != 0) {
            return (double) a / b;
        }
        return 0;
    }
}
```

2.3 Easier Testing and Debugging

Functions can be tested individually, making it easier to find and fix problems.

```java
public static boolean isPrime(int number) {
    if (number <= 1) {
        return false;
    }
    for (int i = 2; i < number; i++) {
        if (number % i == 0) {
            return false;
        }
    }
    return true;
}
```

2.4 Modularity

Functions allow you to break down complex problems into smaller, manageable pieces.

```java
public class StudentGradeSystem {
    public static double calculateAverage(int[] grades) {
        int sum = 0;
        for (int grade : grades) {
            sum += grade;
        }
        return (double) sum / grades.length;
    }

    public static String getLetterGrade(double average) {
        if (average >= 90) return "A";
        else if (average >= 80) return "B";
        else if (average >= 70) return "C";
        else if (average >= 60) return "D";
        else return "F";
    }

    public static void displayResult(String name, double average, String grade) {
        System.out.println("Student: " + name);
        System.out.println("Average: " + average);
        System.out.println("Grade: " + grade);
    }
}
```

3. Practice Questions

Question 1: Basic Function Creation

Write a function called `printWelcome` that takes a person's name as a parameter and prints "Welcome, [name]!" to the console.

Question 2: Mathematical Function

Create a function called `findMaximum` that takes two integers as parameters and returns the larger of the two numbers.

Question 3: String Manipulation

Write a function called `countVowels` that takes a string as input and returns the number of vowels (a, e, i, o, u) in the string.

Question 4: Boolean Function

Create a function called `isLeapYear` that takes a year as an integer parameter and returns true if it's a leap year, false otherwise. (A leap year is divisible by 4, but not by 100, unless it's also divisible by 400)

Question 5: Array Processing

Write a function called `sumArray` that takes an array of integers as a parameter and returns the sum of all elements in the array.

Quiz Questions

1. What is the correct syntax for declaring a function in Java that returns an integer and takes two integer parameters?

a) `function int addNumbers(int a, int b)` b) `public static int addNumbers(int a, int b)` c) `int function addNumbers(a, b)` d) `static int addNumbers(int a; int b)`

Answer: b) public static int addNumbers(int a, int b)

2. What keyword is used when a function doesn't return any value?

a) `null` b) `empty` c) `void` d) `nothing`

Answer: c) void

3. Which of the following is NOT a benefit of using functions?

a) Code reusability
b) Better organization
c) Increased memory usage
d) Easier debugging

Answer: c) Increased memory usage

4. What happens if you try to return a value from a function declared with `void`?

a) The program runs normally b) A compilation error occurs c) The return value is ignored d) The function automatically changes to return type `Object`

Answer: b) A compilation error occurs

5. In the following function declaration: `public static String getName(String first, String last)`, what are `first` and `last` called?

a) Arguments
b) Parameters
c) Variables
d) Return values

Answer: b) Parameters

Practice Coding Questions

1. Temperature Converter Write a function called `celsiusToFahrenheit` that takes a temperature in Celsius as a parameter and returns the equivalent temperature in Fahrenheit. Use the formula: F = (C × 9/5) + 32

2. Password Validator Create a function called `isValidPassword` that takes a string as input and returns true if the password meets these criteria: at least 8 characters long, contains at least one uppercase letter, and contains at least one digit.

3. Grade Calculator Write a function called `calculateGPA` that takes an array of grades (as doubles) and returns the GPA (average of all grades).

4. Prime Number Checker Create a function called `isPrime` that takes an integer as input and returns true if the number is prime, false otherwise.

5. String Reverser Write a function called `reverseString` that takes a string as input and returns the string with its characters in reverse order.